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RESEARCH

Environmental controls on *Gracilariopsis tenuifrons* in mussel farm co-cultivation

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Abstract
Marine co-cultivation offers a promising avenue for sustainable aquaculture; however, its success is driven by a complex interplay of biotic and abiotic factors inherent to the marine environment. Consequently, integrated knowledge is essential for creating effective aquaculture systems. This study aims to understand the environmental factors affecting the red seaweed *Gracilariopsis tenuifrons* cultivated around a mussel farm for biomass production. Seaweeds were cultivated at four locations around a mussel farm (P1, P2, P3, and P4), and at each location, at two depths (surface and 1 m). Environmental parameters (light, nutrients, and circulation) were characterised. P1 and P4 exhibited the highest growth rates, associated with a higher presence of nutrients from the mussel farm brought by the circulation. Depth significantly influenced seaweed growth, with rates at the surface being greater than at deeper depths. Our findings emphasise that a deeper yet accessible understanding of environmental attributes can enhance the eco-efficiency of aquaculture.

Keywords IMTA · Rhodophyta · Seaweed · Environmental knowledge · Eco-efficiency · Circulation pattern

Introduction

Aquaculture production methods are increasingly being optimised to balance sustainability and economic efficiency (Ehrenfeld 2005; Ertör and Ortega-Cerdà 2015; Alexander and Hughes 2017; Biancacci et al. 2022; Nardelli et al. 2023a, b). The mindful use of natural resources, including food, solar energy, water, and aquatic ecosystems, significantly enhances the efficiency of aquatic production. This approach may reduce the need for costly aquaculture inputs and help maintain water quality with less effort, benefiting both the environment and operational efficiency (Chow et al. 2013; Holdt and Edwards 2014; Duan et al. 2020; Lavaud et al. 2020; Turlybek et al. 2025). Integrated Multi-Trophic Aquaculture (IMTA) can meet these goals by being an activity based on recycling resources (Chopin et al. 2004; 2006; Bracken and Stachowicz 2009; Knowler et al. 2020).

IMTA recycles organic and inorganic waste from one trophic level to support another; for example, seaweeds uptake nitrogen and phosphorus from fish and mussel effluents (Buck et al. 2018; Correia et al. 2020; Rusco et al. 2024; Tahiluddin and Roleda 2025). However, the co-cultivation of multiple species in the marine environment is complex and highly influenced by both biotic and abiotic factors (Buck et al. 2018; FAO 2018). For example, the co-cultivation of seaweed and mussel in open water, the capacity of seaweeds to assimilate nutrients from mussel farms is limited by environmental and biological factors (Bannister et al. 2016; Callies et al. 2019). Given the context, the following considerations may be relevant:

1. The temperature and available feed in the seawater (phytoplankton and particulate organic matter) change the mussel's metabolism and, consequently, the farm's rate of inorganic nutrient inputs at sea (Bergström and Mats 2016).
2. Those nutrients are dispersed in the seawater, and the current carries them away, influencing the availability for the seaweed farm, depending on the circulation pattern.

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3. The phytoplankton can compete for nutrients in seawater (Craig et al. 2015).
4. Nitrogenous compounds may be oxidised by microorganisms in seawater with high dissolved oxygen concentration, transforming them less synergistically for seaweed (as ammonium to nitrate) (Newell et al. 2011; Heiss and Fulweiler 2016).
5. The capacity of seaweed to uptake nutrients is controlled by water movement, temperature, and light penetration at the site (Hurd et al. 2014; Roleda and Hurd 2019).

Then, in the end, only a fraction of the nutrients released by mussels can be used by seaweed. These factors can limit the bioremediation potential and biomass production of the seaweed (Broch et al. 2013; Handå et al. 2013; Bannister et al. 2016; Fossberg et al. 2018).

Farmers can enhance co-cultivation by adjusting seaweed depth, which affects light availability. Optimal placement depends on light intensity, water turbidity, and species specific light needs (Nardelli et al. 2019, 2023a; Wilding et al. 2021). Farm design is a controllable variable that can substantially improve the efficiency of seaweed co-cultivation systems. By understanding local hydrodynamic patterns, farmers can optimise the use of mussel metabolic waste by positioning seaweed in areas where seawater currents naturally transport these nutrients, thereby enhancing nutrient uptake and overall system performance (Reid et al. 2020).

Numerical models are increasingly recognised as effective tools for simulating local hydrodynamic patterns and characterising coastal environments (Fossberg et al. 2018). These models enable the interpretation of complex processes occurring within a given area by accounting for the many environmental variables that characterise coastal zones, including waves, winds, and tides (Duarte et al. 2003; Bannister et al. 2016; Hadley et al. 2018). In this context, numerical modelling offers valuable insights into site-specific hydrodynamics, supporting improved cultivation strategies and overall productivity.

This study aimed to provide critical insights into how seaweed growth can be improved using metabolic waste from a mussel farm at sea. Using numerical models, we tested the locations within the mussel farm that could favour the growth of the red seaweed *Gracilariopsis tenuifrons* (C.J. Bird & E.C. Oliveira) Fredericq & Hommersand. Additionally, we tested two depths of seaweed cultivation to determine the optimal cultivation depth.

Material and methods

Seaweed cultivation was integrated into a commercial mussel farm (*Perna perna*; Born, 1778) located in Mar Virado Cove, on the northern coast of São Paulo State, Brazil (23°31′42.70″S, 45°11′20.44″W; Fig. 1). The marine farm is situated in a region with an average depth of 6 m and is exposed to the south-southeast. The surface sediments of the cove are typical of low hydrodynamic environments (Mahiques 1995). The mussel farm consists of a main 100 m longline with droppers spaced every 0.5 m. Mussel biomass per dropper ranges from 10–15 kg. The experiment was conducted during the final stage of mussel cultivation, when individuals had reached adulthood (5 −12 cm).

Experimental cultivation

Gametophytic biomass of *Gracilariopsis tenuifrons* (strain BG0039) was obtained from the Germplasm Bank of the Marine Algae Laboratory 'Edison de José Paula' at the Institute of Biosciences, University of São Paulo. The material was propagated by vegetatively fragmenting 50 g of thallus and cultured under controlled laboratory conditions at 25 ± 1 °C, with a 14:10 h light:dark photoperiod, an irradiance of 100 ± 7 µmol photons m^{-2} s^{-1}, in seawater enriched with von Stosch's solution.

Seaweed cultivation at sea was conducted at four locations (P1–P4) around the marine mussel farm (Fig. 1). Three of these points were positioned along the mussel farm's longline: P1 was located at terminal end of longline, nearest to Lagoinha Beach, P2 was positioned at the midpoint of the longline (~50 m from P1) between the mussel droppers, and P3 was situated at the distal end of the longline, furthest from the shoreline. Additionally, P4 was positioned approximately 20 m inshore from the longline, closer to the coast. P5, located outside the bay approximately 200 m from the mussel farm, served as the control site for seawater nutrient concentration and salinity, and P6 at the Lagoinha River. The experiment was carried out during the spring season, from 29 October to 29 November 2018.

Gracilariopsis tenuifrons was placed in 10 mm nylon tubular nets with a 6 cm diameter, with five individual nets per point ($n = 5$), each containing 5 g of seaweed. Tubular net cultivation was chosen as it has been shown to produce the best growth results for gracilarioid species (Veeragurunathan et al. 2015). Experimental cultivation was conducted at two depths at each point: one at the surface (surface treatment) and the other at a depth of 1 m (depth treatment).

Seaweed biometry

The biometry of *G. tenuifrons* was assessed by the growth rate. Fresh weight was measured weekly over a three-week period (W1, W2, and W3) using a semi-analytical balance with a precision of 0.1 g. After each measurement, the seaweed was returned to the sea without removing any excess biomass. Growth rate (GR) was calculated following Lignell and Pedersén (1989), using the formula:

$$GR\left(\%\ day^{-1}\right) = \left[\left(M_f/M_i\right)^{1/t} - 1\right] \times 100,$$

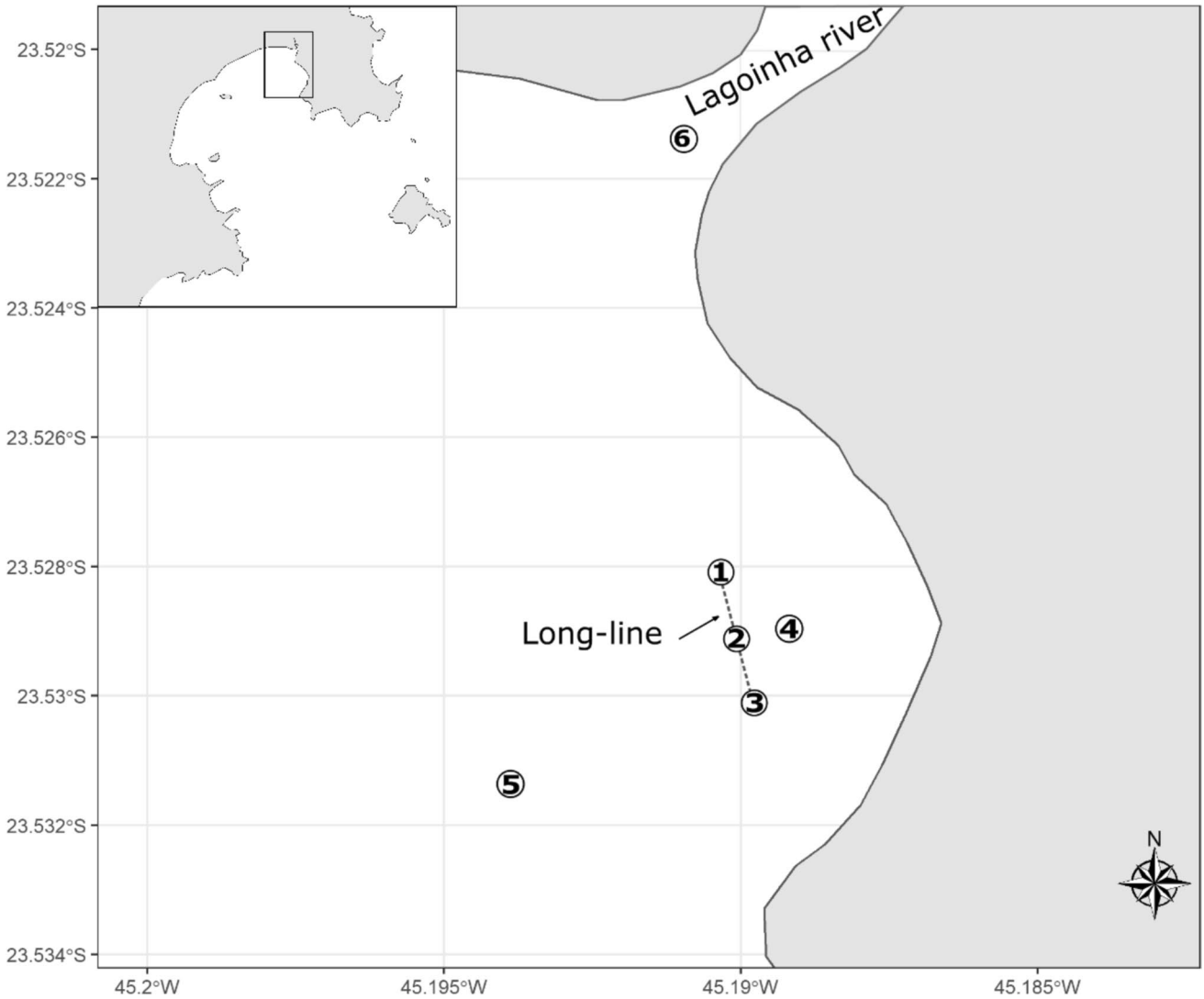


Fig. 1 Map showing Mar Virado Cove on the top left, the polygon shows the area whore the experiment was carried out. Seaweed cultivation was conducted along a mussel farm longline (dotted line) at three points: P1 was located at the end of the longline closest to Lagoinha Beach, P2 at the midpoint of the longline, and P3 at the end furthest from Lagoinha Beach. Additional sampling points included P4, situated approximately 20 m from the longline near the coast; P5, located outside the bay, around 200 m from the mussel farm; and P6, positioned at the Lagoinha River

where M_f is the final mass (g), M_i is the initial mass (g), and t is the number of days between measurements.

Dissolved nutrients and chlorophyll

Seawater samples were collected weekly to characterise nutrient dynamics. Samples were taken from the points where seaweeds were cultivated (P1–P4), as well as from a point outside the mussel farm (P5), to monitor nutrient variations in an area unaffected by the presence of the mussel farm (Fig. 1), and from the Lagoinha River (P6), as river discharge could influence nutrient input or dilution.

Seawater samples were kept at 4 °C in an icebox and transported to the "Clarimundo de Jesus" Research Base of the Oceanographic Institute, University of São Paulo. In the laboratory, the samples were filtered through 0.45 µm Whatman GF/F membranes. Material retained on the filters was transferred to polypropylene tubes and stored at −20 °C, protected from light, until chlorophyll content analysis. The filtered seawater was collected in 250-mL plastic bottles and stored at −20 °C for subsequent analysis of dissolved nutrients. Sample analysis was conducted approximately one month after the experiment concluded.

The analysis of dissolved nutrients and chlorophyll in seawater was conducted at the Laboratory of Biogeochemistry

of Nutrients, Micronutrients, and Trace Elements at the Oceanographic Institute, University of São Paulo. Urea, ammoniacal nitrogen (ammonium + ammonia), phosphate, silicate, and chlorophyll concentrations were determined using colourimetric methods on a Thermo Evolution 201 spectrophotometer.

Urea content was evaluated following the method described by Aminot and Kerouel (1982), with adaptations by Braga (pers. comm.), using a wavelength of 526 nm and an accuracy of ± 0.15 μmol L^{-1}. Ammoniacal nitrogen concentration was determined according to Tréguer and Le Corre (1975) at a wavelength of 630 nm, with an accuracy of ± 0.05 μmol L^{-1}. For nitrate determination, nitrate was first reduced to nitrite using a copperised cadmium column as described by Tréguer and Corre (1975). The concentrations of reduced nitrate and nitrite were then measured using a BranLuebbe AutoAnalyzer II system in a segmented flow, following methods outlined in the following references (Wood et al. 1967; Tréguer and Corre 1975; Braga 1997a, b; Grasshof et al. 1999), with an accuracy of ± 0.01 μmol L^{-1} for nitrite and ± 0.02 μmol L^{-1} for nitrate. Nitrate concentration was calculated by subtracting the measured nitrite concentration from the total nitrite content obtained after nitrate reduction (NO_2^- + reduced NO_3^-). Phosphate concentration was assessed following the method described in Grasshof et al. (1999) using a wavelength of 880 nm, quartz cuvettes with a 50 mm optical path, and an accuracy of ± 0.01 μmol L^{-1}. Silicate concentration was determined according to Grasshof et al. (1999) using 10 mm optical path cuvettes at a wavelength of 810 nm, with an accuracy of ± 0.05 μmol L^{-1}. Active chlorophyll retained on the filters was extracted in 90% acetone for 24 h at 4 °C and chlorophyll concentrations were determined following the method of Strickland and Parsons (1968).

Climatological data

Temperature and solar irradiance were measured using HOBO Pendant Temperature/Light Data Logger probes, positioned at the surface and at 1 m depth at point P1 (Fig. 1). Data were collected on an hourly basis. Salinity was measured with an Atago ATC-S/MIII refractometer at points P1 to P6 (Fig. 1), with samples collected during the tidal slack.

River discharge

Near the marine farm, approximately 1 km from the cultivation area, there is an input from the Lagoinha River (Fig. 1; P6). Although it is a small river, it can influence the cultivation points; therefore, its flow was estimated weekly. A longitudinal river section near the mouth was defined with three sampling points, where river width and depth were measured to calculate the cross-sectional area. The river flow was then estimated by multiplying this area by the water velocity. River velocity was measured using the STRAVA mobile app, which tracked location over time to calculate average travel speed. A mobile phone, sealed in a waterproof bag, was allowed to drift with the river current over a distance of approximately 50 m, and STRAVA recorded the data. This procedure was repeated three times, and the average velocity was calculated to estimate river flow. The resulting measurements were incorporated into the hydrodynamic model described below.

Hydrodynamical model

The circulation pattern of the study area was modelled in collaboration with the Coastal Dynamics Laboratory of the Oceanographic Institute, University of São Paulo, using the Delft3D software developed by the independent research institute Deltares (Netherlands). Delft3D is a modular system, with the hydrodynamic component (Delft3D-Flow) being particularly suited for studies of hydrodynamics and pollutant dispersion (Deltares 2014). The model domain encompassed the study area, and the computational grid was generated using the RGF-Grid support module. Bathymetry and coastline data were sourced from the Coastal Dynamics Laboratory database.

The Delft3D-Flow module simulates hydrodynamic and multidimensional transport (2D or 3D) by calculating non-stationary flows influenced by meteorological and astronomical (tidal) forces within both curved and regular grid contours. The model is based on the Navier–Stokes equations, adapted for shallow water conditions. Bathymetric contours were interpolated across the grid, and input data included tidal forcing (from Delft Dashboard) and wind direction and intensity (from the meteorological station of the Oceanographic Institute at Ubatuba, located approximately 8 km from the study site).

The model parameters considered water density (1027 kg m^{-3}) and gravitational acceleration (9.8 m s^{-2}). To calibrate the model, an Acoustic Doppler Current Profiler (ADCP; Aquadopp Nortek 1 MHz) was deployed near the marine farm (23°31′45.25" S, 45°11′21.14" W) to collect current velocity and direction data. The ADCP was configured with a sampling interval of 30 min and a vertical cell size of 1 m, operating continuously for 30 days to cover a complete tidal cycle and spring-neap tidal variations.

Model calibration and validation were first performed qualitatively, followed by quantitative evaluation using the Mean Relative Absolute Error (MRAE) (Table 1), following Van Rijn et al. (2000), and the Skill parameter (Table 1), as described by Warner et al. (2005).

Statistical analysis

The results were first tested for normality using the Shapiro–Wilk test. A three-way analysis of variance (ANOVA)

was then applied to assess growth rate differences across sample points, with time, location, and depth as fixed factors. Nutrient and chlorophyll concentrations in seawater were analysed using one-way ANOVA. When significant differences were detected, Tukey's post-hoc test was applied for pairwise comparisons. Statistical significance was considered at $p < 0.05$ (Zar 1996). Principal Component Analysis (PCA) was used to explore the relationships among environmental parameters at the cultivation points and their influence on seaweed development. All statistical analyses were performed using RStudio version 4.2.2.

Results

P1 and P4 demonstrated the highest growth rates throughout the experiment, a pattern that remained consistent each week (Fig. 2). Growth peaked during the first week across all cultivation points, while the second and third weeks showed similar, more moderate rates. Depth also played a significant role, with seaweeds cultivated at the surface achieving higher growth in all weeks (Fig. 2A, B; significant week × point × depth interaction, Table 2). The average growth rate throughout the experiment was for the surface (P1 8.0 ± 0.6; P2 5.25 ± 1.0, P3 4.34 ± 0.7 and P4 8.5 ± 0.9 (% day^{-1})), bottom (P1 6.1 ± 0.6; P2 4.7 ± 0.5, P3 3.7 ± 0.9 and P4 6.7 ± 1.1 (% day^{-1})). Following the growth rate, the biomass production was significantly higher at P1 and P4 than at P2 and P3 (Fig. 2C, D; significant week × point × depth interaction, Table 2).

No substantial temperature fluctuations occurred over the course of the experiment. Surface temperatures ranged from 23.0 to 26.5 °C, while depth temperatures varied between 22.9 and 25.3 °C, and statistical analysis detected no significant differences between sampling points or depths (ANOVA $F = 0.901$, $df = 41.81$, $p = 0.449$). Daily average solar irradiance was higher in weeks 1 and 3 than in week 2 (Fig. 3). At the surface, irradiance averaged 206.98 ± 133.45 µmol photons m^{-2} s^{-1} in week 1, dropped to 104.13 ± 96.49 µmol photons m^{-2} s^{-1} in week 2, and then rose to 209.13 ± 138.90 µmol photons m^{-2} s^{-1} in week 3. Corresponding depth values were 98.03 ± 61.85, 32.01 ± 44.59, and 82.62 ± 68.50 µmol photons m^{-2} s^{-1}, respectively. Although weeks 1 and 3 did not differ significantly from each other, irradiance in week 2 was markedly lower (ANOVA $F = 30.218$, $df = 840.43$, $p < 0.001$). Salinity remained around 33–34 ppm, with no significant difference (ANOVA $F = 0.026$, $df = 7.473$, $p = 0.99$).

Seawater nutrient concentrations and chlorophyll-*a* levels at the four cultivation points and P5 (Table 3). Ammoniacal nitrogen (NH_4^+–N) was significantly elevated at P1 relative to all other points, with P2, P3 and P4 also exhibiting higher values than P5. Both nitrite (NO_2^-) and nitrate (NO_3^-) concentrations were greatest at P1 and P4 compared with the remaining points. In contrast, phosphate (PO_4^{3-}), silicate (SiO_2), chlorophyll-*a*, and urea did not differ significantly among any of the sampling locations. At sampling site P6, located at the mouth of the Lagoinha River, the concentrations of dissolved inorganic nutrients and chlorophyll-*a* were as follows (µmol L^{-1}): NH_4^+–N: 7.1 ± 2.3; NO_2^-: 0.65 ± 0.09; NO_3^-: 10.1 ± 1.3; PO_4^{3-}: 0.24 ± 0.05; SiO_2: 62.1 ± 6.4; chlorophyll-*a*: 0.41 ± 0.05; and urea: 0.45 ± 0.2.

A principal component analysis (PCA; Fig. 4) incorporating nutrient concentrations (excluding silicate), chlorophyll-a, urea, and growth rates revealed two distinct clusters. The first two components accounted for 67.07% of the total variance (PC1 = 46.60%, PC2 = 20.47%). Points P1 and P4 clustered together, driven primarily by high phosphate, nitrite, nitrate and growth-rate loadings, while P2 and P3 formed a separate group. Points P1, P2, and P3 were predominantly driven by elevated ammoniacal nitrogen, whereas urea played only a minor role in cluster separation. Additionally, chlorophyll-*a* displayed a strong inverse relationship with the nutrient levels.

The numerical model was made using the variation in sea level, and validation was tested. In the quantitative evaluation, the value received in MRAE was 0.476 (classified as reasonable) and for Skill 0.930 (classified as excellent) (see Table 1). The circulation pattern in the study area can be seen in Fig. 5, representing an area with a low average flow, not exceeding 1.2 cm s^{-1}. The preferred circulation is clockwise, and the current enters parallel to the long-line cultivation system from P3 to P1 (Fig. 5A). The current direction changes clockwise from the direction of the long-line to P4 (Fig. 5B), then the flow reverses, at low tide, in a direction almost parallel to the long-line, going from P4 to the cultivation of mussels (Fig. 5C, D). It can be noted that the flow velocity is always lower in the area close to P1 and P4. The Lagoinha River's direct influence on cultivation was also verified. One hundred drifters were launched at the river's mouth in the second week of the experiment when the river flow was highest. Figure 6 shows that 87% of drifters travelled westward, away from the farm area, while the remaining 13% that remained near the crop took approximately one week to arrive.

Table 1 Mean Relative Absolute Error (MRAE) classification by Van Rijn et al. (2003) and Skill classification by Henriksen et al. (2003)

MRAE Classification		Skill Classification	
Excellent	< 0.1	Excellent	> 0.85
Good	0.1–0.3	Good	0.65–0.85
Reasonable	0.3–0.5	Reasonable	0.5–0.65
Poor	0.5–0.7	Poor	0.2–0.5
Bad	> 0.7	Very Bad	< 0.2

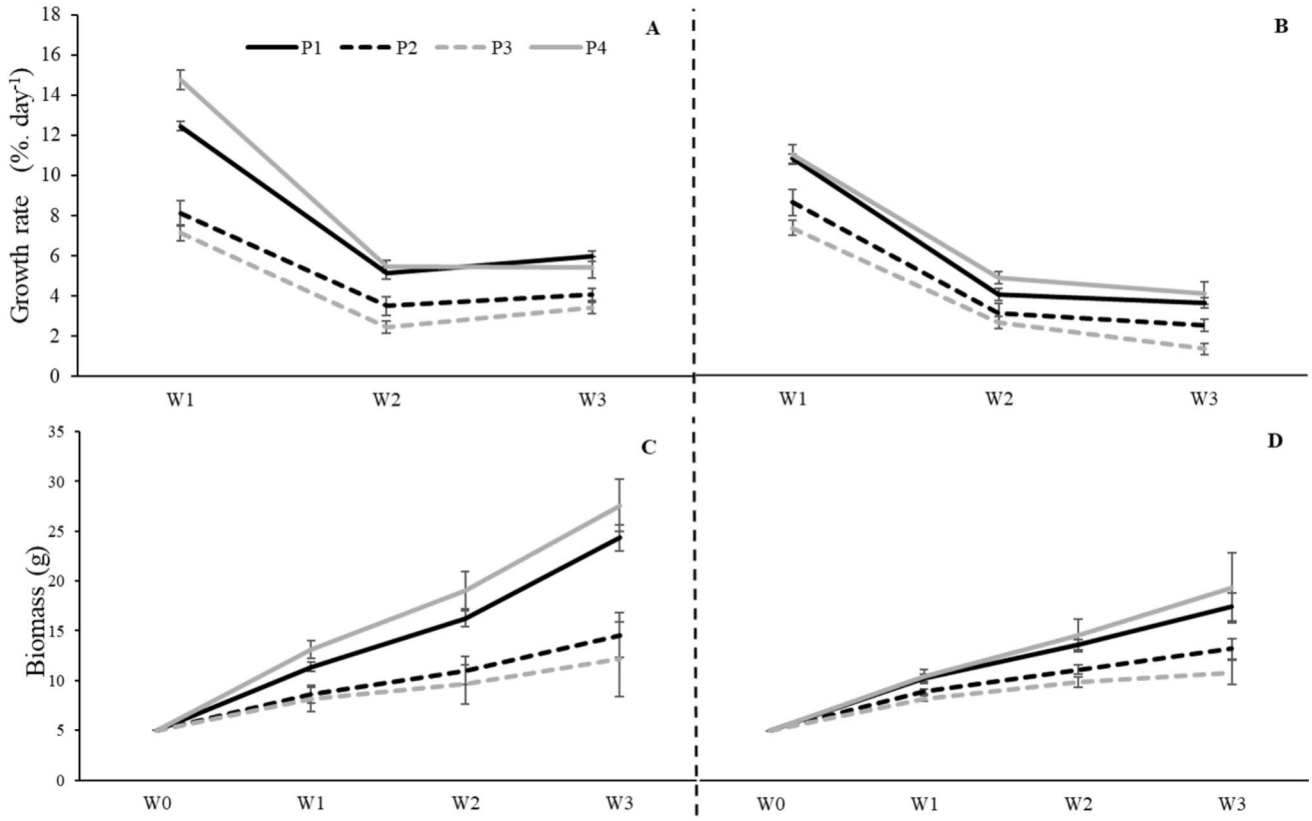


Fig. 2 Growth rates (mean ± S.D., $n = 5$) of *G. tenuifrons* over the weeks cultivated associated with the mussel farm at points P1, P2, P3, and P4. (**A**) Seaweeds grew at the surface. (**B**) Seaweed grew to a depth of 1 m. (**C**) Seaweed biomass at the surface. (**D**) Seaweed biomass to a depth of 1 m

Table 2 Statistical results from three-way ANOVA (point, week and depths) for growth rate and biomass

	Growth rate			Biomass		
	F	*df*		*F*	*df*	*p*
Point	87.02	205.6	< 0.001	66.24	509.9	< 0.001
Week	579.72	913.2	< 0.001	227.59	1168.0	< 0.001
Depth	54.97	43.3	< 0.001	154.89	397.5	< 0.001
Point × Week	6.25	29.5	< 0.001	10.38	159.9	< 0.001
Point × Depth	3.68	8.7	0.014	121.5	15.78	0.014
Week × Depth	6.71	10.6	< 0.001	78.0	15.19	< 0.001
Point × Week × Depth	7.45	35.2	< 0.001	0.635	9.8	0.702

Discussion

Successful seaweed cultivation depends on understanding the conditions necessary for its growth and development. In land-based systems, these conditions (e.g., temperature, light, nutrients) can be relatively easily controlled (Hafting et al. 2012; FAO 2018; Nardelli et al. 2019, 2023a; Khan et al. 2024). At sea, control over environmental conditions is highly limited; therefore, a comprehensive understanding of the site's biotic and abiotic parameters is critical for optimising seaweed growth and productivity (Bracken and Stachowicz 2009; Perini and Bracken 2014; Buck et al. 2017, 2018; Roleda and Hurd 2019; Giangrande et al. 2021).

Light

Light intensity decreases with depth, and in some locations, pronounced attenuation may occur immediately below the surface. Therefore, characterising the vertical light attenuation profile is instrumental for understanding the underwater

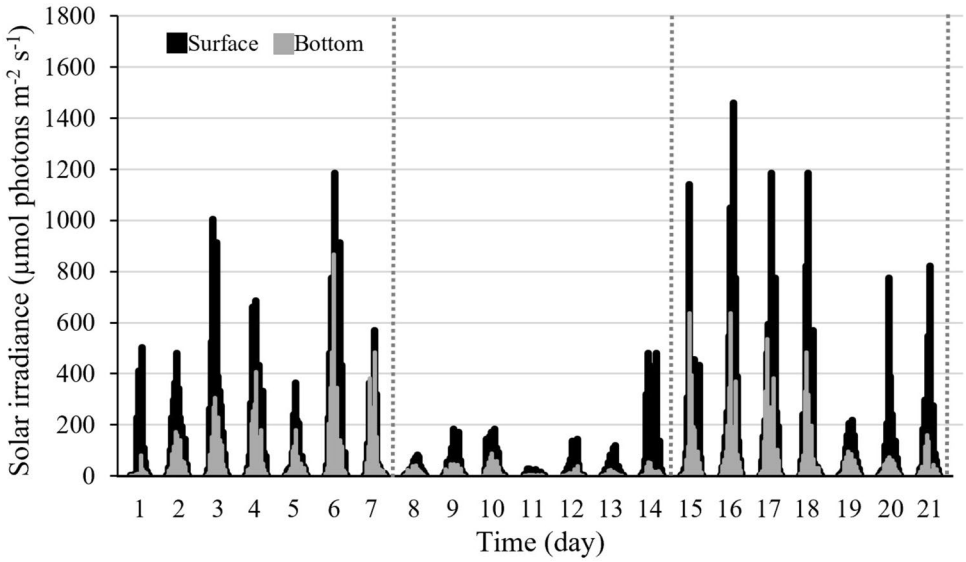


Fig. 3 Solar irradiance at the surface and 1 m depth over the three weeks of cultivating *G. tenuifrons* associated with a mussel farm

Table 3 Seawater nutrients and chlorophyll-*a* (µmol L^{-1}) at different long-line points of the mussel farm and reference points

	P1	P2	P3	P4	P5	*F*	*df*	*p*
N-ammoniacal	0.43 ± 0.05^{a}	0.29 ± 0.06^{b}	0.30 ± 0.06^{b}	0.21 ± 0.05^{b}	0.09 ± 0.02^{c}	15.91	4.58	< 0.001
Nitrite	0.10 ± 0.02^{ab}	0.08 ± 0.02^{bc}	0.07 ± 0.01^{bc}	0.11 ± 0.01^{a}	0.07 ± 0.01^{c}	10.81	4.03	< 0.001
Nitrate	0.39 ± 0.08^{a}	0.29 ± 0.06^{b}	0.26 ± 0.05^{b}	0.41 ± 0.1^{a}	0.29 ± 0.07^{b}	9.62	4.41	< 0.001
Phosphate	0.40 ± 0.03^{a}	0.37 ± 0.05^{b}	0.39 ± 0.06^{ab}	0.43 ± 0.04^{a}	0.34 ± 0.05^{c}	8.451	4.01	< 0.001
Silicate	10.51 ± 1.51	11.00 ± 1.57	10.05 ± 1.80	10.24 ± 1.42	9.98 ± 2.02	0.434	5.49	0.783
Chlorophyll-*a*	2.62 ± 0.61	2.58 ± 0.60	2.80 ± 0.58	2.67 ± 0.60	2.91 ± 1.22	0.164	4.97	0.955
Urea	0.21 ± 0.04	0.15 ± 0.03	0.20 ± 0.06	0.26 ± 0.07	0.16 ± 0.02	2.74	4.02	0.054

P1: placed at the end of the long-line towards Lagoinha Beach, P2: placed at the center of the long-line, P3: placed at the opposite end of Lagoinha Beach, P4: outside the long-line, approximately 20 m distant from the long-line and close to the coast, P5: outside the bay and 200 m from the mussel farm. Statistical differences are represented as lowercase letters

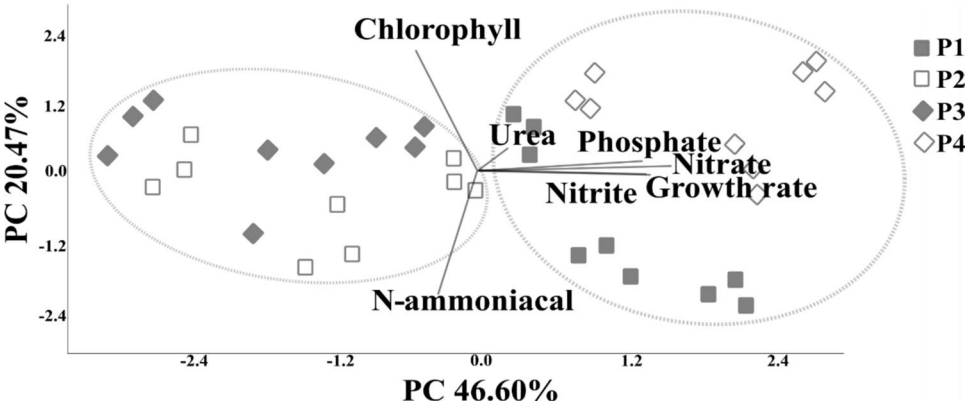
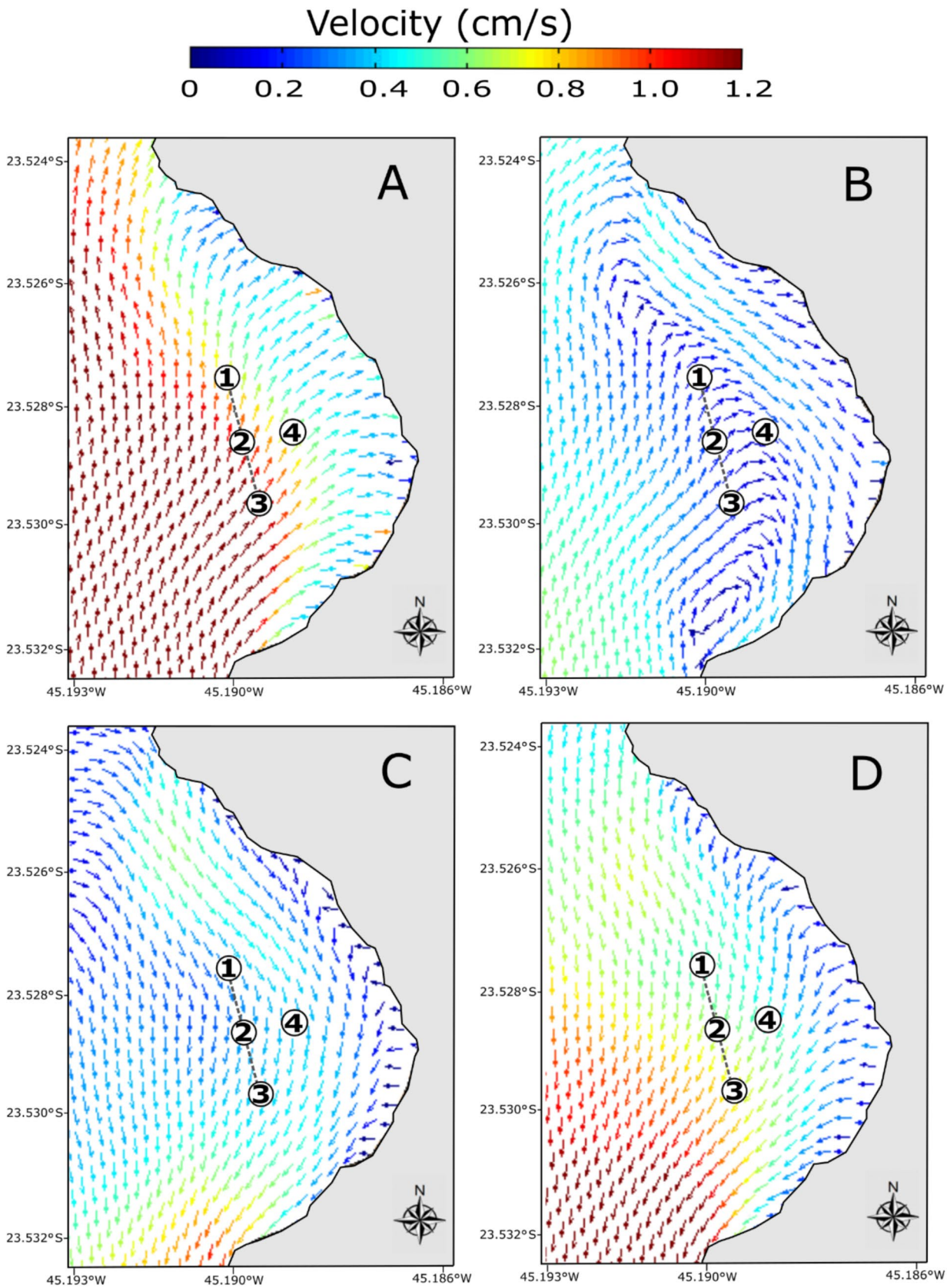


Fig. 4 Analysis of the main components (PCA) of seaweed cultivation (P1, P2, P3, and P4), under the influence of inorganic nutrients (except silicate), urea, chlorophyll-*a*, and growth rate

light environment relevant to seaweed cultivation (Morgan et al. 2020). In this study, the mean surface irradiance was 189 µmol photons m^{-2} s^{-1}, while at a 1 m depth, it declined to 78 µmol photons m^{-2} s^{-1}, representing an approximate 60% reduction. This reduction in light availability appears to have influenced the development of *G. tenuifrons*, as consistently higher growth rates were observed at the surface across all sampling points and throughout the cultivation period. This finding is consistent with a previous laboratory-based study where Torres et al. (2015) reported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◂**Fig. 5** The circulation pattern in the spring tide of the study area. The preferred circulation is clockwise, (**A**) the current enters parallel to the long-line cultivation system in the direction P3 to P1, (**B**) the current direction changes clockwise from the direction of the long-line to P4; (**C** and **D**) the flow is reversed, at low tide, in a direction almost parallel to the long-line, going from P4 to the cultivation of mussels

that *G. tenuifrons* exhibited significantly higher growth rates under high irradiance, reaching 12.6 ± 1.75% day^{-1} at 1000 µmol photons m^{-2} s^{-1}, compared to 6.40 ± 0.8% day^{-1} at 60 µmol photons m^{-2} s^{-1}. Other seaweed species exhibit similar responses, with reduced growth observed under lower irradiance conditions at greater depths: *Gracilaria bursa-pastoris* (S.G.Gmelin) P.C.Silva had a growth rate of 4.95 ± 0.29% day^{-1}, and it did not exceed 2% day^{-1} at 4 m depth (Marinho-Soreano 2012); *Hypnea musciformis* (Wulfen) J.V.Lamouroux had 7.0 ± 0.6% day^{-1} at the surface and 1.3 ± 0.6% day^{-1} at 1.2 m (Ganesan et al. 2006); *Kappaphycus alvarezii* (Doty) Doty ex P. C. Silva had 6.8 ± 0.5% day^{-1} on the surface and 5.2 ± 0.2% day^{-1} at a depth of 1 m (Hayashi et al. 2008). On the other hand, *Lessonia corrugata* A.H.S.Lucas shows better growth at 3 m depth (7.9 ± 0.23% day^{-1}) than at 1 m depth (4.6 ± 0.7% day^{-1}) (Nardelli et al. 2023b). This study emphasises the significance of integrating knowledge of seaweed irradiance requirements with the vertical distribution of light availability in the marine environment (Rule et al. 2025). Understanding it can support seaweed cultivation in response to seasonal fluctuations in irradiance. For instance, if a species grows optimally at a depth of 3 m in spring, cultivating it at a shallower depth during winter can reduce the impact of lower light levels in that season.

Growth rates varied markedly across the weeks, likely influenced by light availability and seaweed density. The highest growth rate coincided with both elevated light intensity and low initial seaweed density in the first week, conditions that likely favoured rapid growth. In the second week, average irradiace declined sharply, corresponding with a notable reduction in growth rate. Although light intensity increased again in the third week, growth did not recover proportionally. This attenuated growth response may be attributed to the increased standing biomass, where higher density likely induced self-shading and intensified intraspecific competition, thereby limiting resource availability and constraining growth relative to the first week (Xiao et al. 2019; Ross et al. 2024).

The river's nutrient input

Nutrient delivery to seaweed farms in coastal areas can be considerably affected by fluvial inputs (Berger et al. 2022). This influence depends on river discharge, local hydrodynamics, and proximity of the farm to the river mouth. In our study area, the river mouth was located over 1 km from the seaweed farm. The dominant direction of the riverine flow was westward, away from the farm, with only a minor fraction of the discharge transported towards the cultivation site (Fig. 6). The inflow of riverine water into the vicinity of the seaweed farm was slow, requiring approximately one week to reach the cultivation area. This gradual movement facilitated extensive mixing between freshwater and seawater by the action of waves, wind, and tide (Frigstad et al. 2020; Berger et al. 2022). Salinity and silicate concentrations, both of which are indicators of fluvial influence and the mixing process, did not show significant variation between the cultivation points, suggesting a limited and uniform impact of the river discharge across the area (Morris et al. 1978; Svendsen and Putrevu 1996; Ragueneau et al. 2005; Bastos and Braga 2018).

Mussel nutrient input

Mussel aquaculture is a relatively low-footprint activity, mainly because production relies entirely on the filter-feeding activity of the mussels, rather than on externally supplied feed (Gibbs 2004; Ferreira et al. 2007). Consequently, the nutrient load on the surrounding seawater is low compared to fish or shrimp farms, where feed inputs can trigger seawater eutrophication (Buck et al. 2017). The nutrient input can be reduced by integrating mussel farming with seaweed. The seaweed uptakes excess nutrients from the surrounding seawater, promoting its growth while decreasing the chance of eutrophication (Gibbs 2004; Jansen et al. 2012). Several studies have demonstrated that mussel farms can positively influence seaweed growth through nutrient input (Visch et al. 2020; Hargrave et al. 2021, 2022). However, for seaweed to effectively utilise available nutrients, those nutrients must be transported to the seaweeds, making a thorough understanding of the hydrodynamics within the cultivation area essential.

In our study, *G. tenuifrons* consistently exhibited higher growth rates at the two points within the mussel farm (P1 and P4), with mean growth rates approximately 70% greater than those observed at the points P2 and P3. Based on the hydrodynamic model and nutrient concentrations measured in seawater samples, points P1 and P4 received higher nutrient inputs attributable to the metabolic excretion of mussels during the study period (Table 3; Fig. 5).

Mussels release nutrients via excretion and biodeposition after filtering organic matter from seawater. According to Pessatti et al. (2002), an individual *Perna perna* mussel can filter up to 1.04 L h^{-1} of seawater. Based on this filtration rate, the mussel farm used in this study is estimated to filter approximately 780,000 L h^{-1}. Our data shows a slight

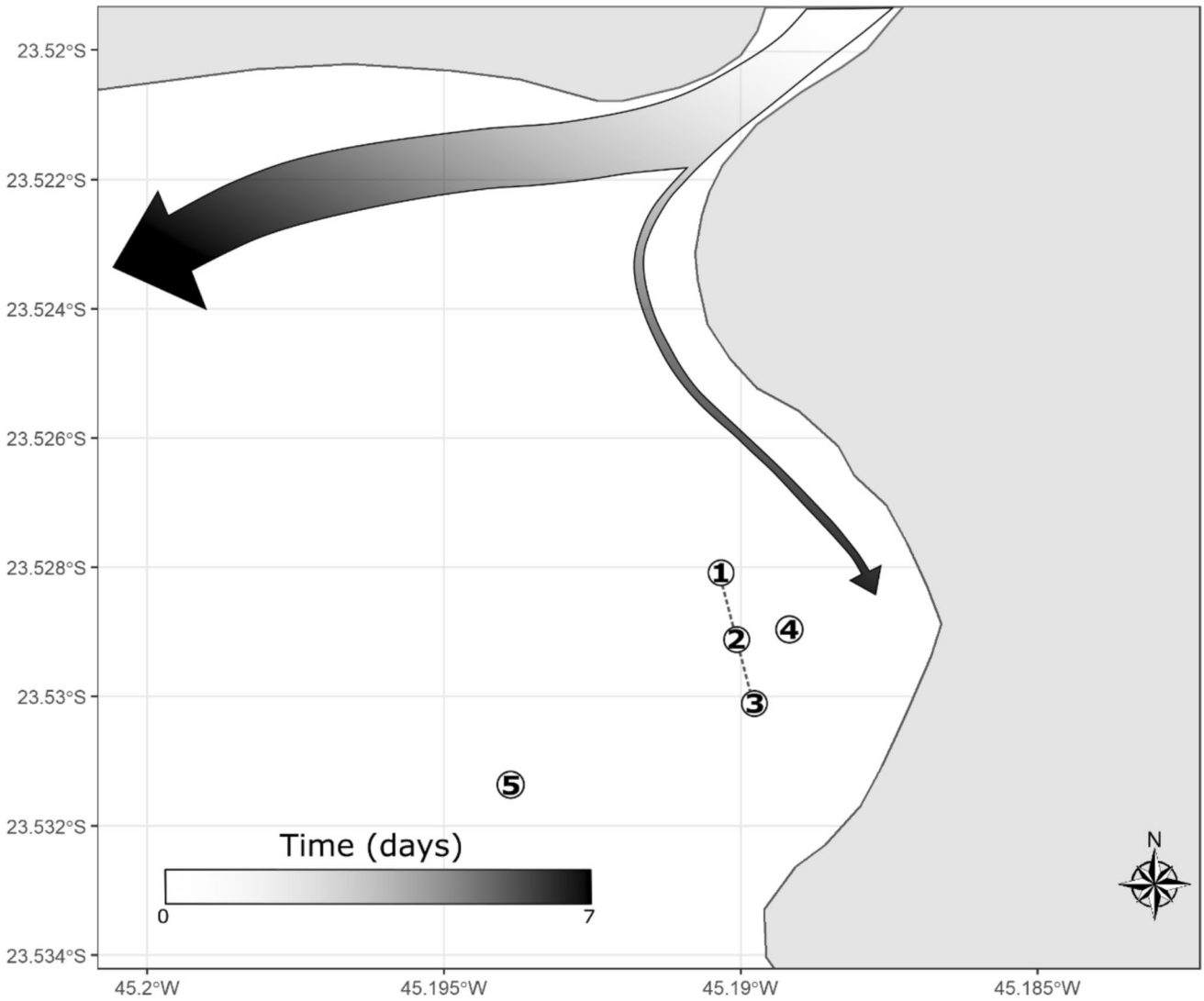

Fig. 6 Path taken by the drifters that were launched at the mouth of the Lagoinha River. A large number of drifters go west, parallel to Lagoinha beach. A small fraction of the drifters go in the direction of cultivation, taking a week to reach the area of cultivation

reduction in chlorophyll-a concentrations at points P1 and P4, indicating that phytoplankton consumption may be occurring due to the presence of the mussel farm. This aligns with increased nutrient levels at these points and supports our hydrodynamic model.

N-ammoniacal concentrations were highest at long-line points, especially P1. Due to mussel excretion, it may contain up to 70% N-ammoniacal (Jansen et al. 2012). Less oxidised nitrogen forms, such as ammoniacal nitrogen, are more readily assimilated into organic compounds (Zheng et al. 2017). Seaweeds typically uptake nitrate, which requires energy-intensive enzymatic reduction to ammonium via nitrate and nitrite reductases (Haines and Wheeler 1978; Chow et al. 2013). In contrast, environments enriched with ammoniacal nitrogen enhance nutrient uptake and promote seaweed growth (Bigelow 1978; Nardelli et al. 2019). Consequently, the increased growth observed in P1 probably resulted from a significant influence of ammonia nitrogen availability. Several studies have demonstrated a preferential uptake of ammoniacal nitrogen (NH_4^+) in rhodophytes, including *Gracilaria tikvahiae* (McLachlan & Bird, 1984), *Gracilaria pacifica* (Haynes & Druehl, 1994), *Gracilaria gracilis* (Smit 2002), and *Gracilaria vermiculophylla* (Abreu et al. 2011).

The higher nitrate and nitrite concentrations may also be linked to the growth in P4 and P1. The observed enrichment may be attributed to the microbial oxidation of ammonium (Beman et al. 2002). Several studies have reported ammonium nitrification rates in coastal regions, including 0.001 μmol L^{-1} h^{-1} (Newell et al. 2011), 0.002–0.005 μmol L^{-1} h^{-1} (Zheng et al. 2017), and 0.5 μmol m^{-2} h^{-1} (Heiss and Fulweiler 2016). These rates indicate the rapid oxidative

transformation of nitrogen compounds in coastal environments, likely supported by adequate dissolved oxygen availability. Higher phosphate levels at P1 and P4 likely result from preferential transport. This nutrient is essential for seaweed growth, supporting ATP-driven energy transfer in photosynthesis and respiration (Lee et al. 2005; Marinho-Soriano et al. 2009). No correlation was found between seaweed growth and urea concentrations at cultivation points, leaving uncertainty about whether urea is assimilated directly or via microbial degradation (Tyler et al. 2005; Smith et al. 2018).

Although the study area is oligotrophic, an environmental condition typically unfavourable for seaweed cultivation (Ateweberhan et al. 2015), *G. tenuifrons* exhibited a high growth rate of approximately 15% day^{-1} under integrated culture conditions. This value surpasses previously reported growth rates for *G. tenuifrons*. Gómez and Millán (1997) observed 11.54 % day^{-1} at the same period of the year, on Margarita Island, Venezuela (Gómez and Millán 1997), and Rincones and Moreno (2011) found 0.59 ± 0.39 % day^{-1} (90 day average) on the Guajira Peninsula, Colombia (Rincones and Moreno 2011). Enhanced performance under integrated aquaculture has also been observed in other rhodophyte species. For instance, Marinho-Soriano et al. (2009) reported that *Gracilaria birdiae* E.M.Plastino & E.C.Oliveira can retain over 90% of nutrients derived from shrimp farming effluents. Similarly, *Gracilaria lemaneiformis* (Bory) Greville has demonstrated efficient uptake of phosphorus and nitrogen from fish farm effluents in open-water cultivation systems (Duan et al. 2019).

Conclusion

The integrated cultivation of the mussel *Perna perna* and the seaweed *Gracilariopsis tenuifrons* showed potential to enhance seaweed biomass production. *G. tenuifrons* exhibited greater growth near the surface, indicating a possible preference for higher solar irradiance levels characteristic of the studied points and season. Higher growth rates were observed at cultivation sites downstream of mussel farms, where nutrient input from mussel excretion may be elevated. This integrated system may provide environmental benefits by enabling the seaweed to assimilate nutrients derived from the mussels' nitrogen metabolism. Consequently, seaweed biomass production could contribute to mitigating excess eutrophic compounds associated with mussel farming.

However, it's important to highlight the limitations of this study. We were only able to conduct a cultivation cycle for a specific period of one month. Our findings reflect only a small period of time in the cultivation cycle, and further studies for other periods of the year are necessary. Furthermore, for better resolution and comparison with circulation models, increasing the number of cultivation points would enhance the resolution in future studies. However, the main finding is that multiple factors must be considered for integrated offshore cultivation to better utilise resources.

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Authors' contributions AN: designed the study; experimental work, data analysis, wrote the initial draft of the manuscript, provided feedback on previous versions of the manuscript and approved the final draft; LCA: data analysis, provided feedback on previous versions of the manuscript and approved the final draft; PS: data analysis, provided feedback on previous versions of the manuscript and approved the final draft; VGC: data analysis, feedback on the experimental design and results, provided feedback on previous versions of the manuscript, and approved the final draft; ES: data analysis, feedback on the experimental design and results, provided feedback on previous versions of the manuscript, and approved the final draft; ESB: designed the study, provided feedback on previous versions of the manuscript and approved the final draft; VG: designed the study, provided feedback on previous versions of the manuscript and approved the final draft; FC: designed the study, provided feedback on previous versions of the manuscript and approved the final draft.

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Declarations

Competing interests The authors declare no competing interests.

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